

Do Register Tokens Regularize Vision Transformers Under Data Scarcity?

A controlled 12-run ablation on low-data CIFAR-100

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Registers at scale. Darcet et al. (ICLR 2024): large ViTs recycle a few patch tokens as scratch space. Adding K learnable *register* tokens removes the artifact.

That was shown where there is *too much* data to overfit.

Our question

When a ViT is starved of data, do registers reduce the generalization gap — and does any such reduction transfer to held-out performance?

H1 Structural regularization: $\Delta\mathcal{L}$ falls.

H2 Capacity dilution: large K hurts at $d = 192$.

H3 Registers prevent attention-entropy collapse.

Matrix. $K \in \{0, 1, 4, 8\} \times \text{seeds}$
 $\{42, 1337, 3407\}$.

Data. CIFAR-100, 100 images/class: 9,000 train / 1,000 val, plus the untouched 10,000-image test split.

Model. ViT-Tiny, $d = 192$, 12 blocks, ImageNet-pretrained. $K = 8$ adds 1,536 of 5.54 M parameters.

Budget. 50 epochs, AdamW, cosine, AMP. 2.83 GPU-hours total.

What makes it a controlled test

Within a seed, the subset, the split, the batch order and every hyperparameter are identical across arms. K is the only thing that moves.

Why paired statistics

At $n = 3$, between-seed spread (2.17 pp) swamps any treatment effect. All differences are formed *within* a seed, then reduced.

Result 1: no arm improves held-out Top-1

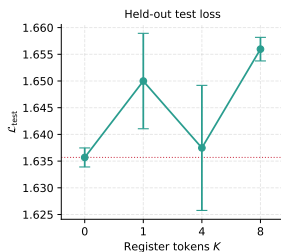
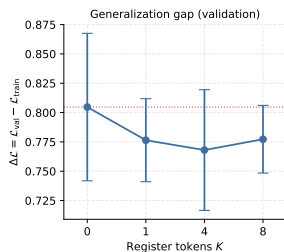
Configuration	Test Top-1 (%)	Test Loss	Val Loss	Gap $\Delta\mathcal{L}$
Baseline ($K = 0$)	75.19 $\pm$ 0.24	1.6357 $\pm$ 0.0018	1.6829 $\pm$ 0.0609	0.8047 $\pm$ 0.0628
Registers ($K = 1$)	74.66 $\pm$ 0.42	1.6500 $\pm$ 0.0089	1.6545 $\pm$ 0.0380	0.7764 $\pm$ 0.0354
Registers ($K = 4$)	75.06 $\pm$ 0.33	1.6375 $\pm$ 0.0117	1.6492 $\pm$ 0.0536	0.7680 $\pm$ 0.0514
Registers ($K = 8$)	74.67 $\pm$ 0.12	1.6560 $\pm$ 0.0022	1.6602 $\pm$ 0.0268	0.7773 $\pm$ 0.0288

The bolding splits down the middle: the control wins both test columns, $K = 4$ wins both validation-derived columns.

Paired per seed: $K = 1$ and $K = 8$ lose accuracy in **3 of 3** seeds; $K = 4$ wins in 1 of 3 ($\bar{\delta} = -0.13$ pp, $p = 0.466$).

Top-5 is the one place a register arm leads — $K = 4$ at 92.54 ± 0.23 vs 92.43 ± 0.14 — but paired it is $+0.11$ pp with SD 0.39, 2/3 seeds, $p = 0.681$.

Result 2: H1 holds on validation and dies on test



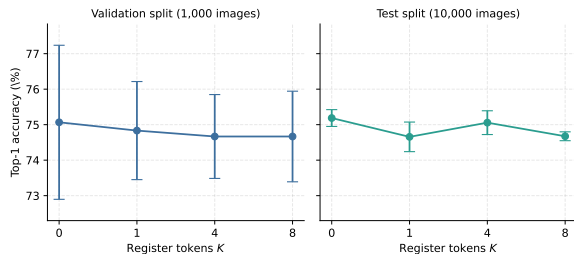
$K = 4$ vs. control, paired over seeds:

	$\bar{\delta}$	Dir.	p
Val loss	-0.0337	3/3	0.024
Gen. gap	-0.0367	3/3	0.051
Test loss	+0.0018	2/3	0.823

Why it dies: final training loss is 0.8782 / 0.8780 / 0.8812 / 0.8829 across $K = 0, 1, 4, 8$ — a spread of 0.005.

A regularizer trades training fit for held-out fit. Nothing was traded.

Why the validation split misled us



Between-seed spread of Top-1,
control arm:

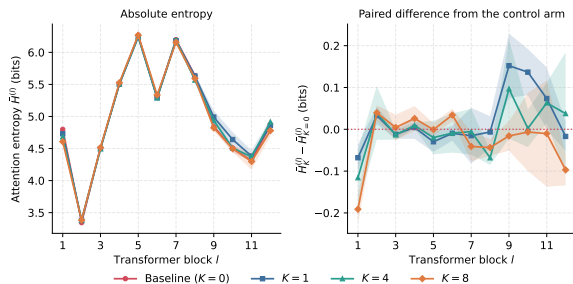
- ▶ validation (1,000 images): **2.17 pp**
- ▶ test (10,000 images): **0.24 pp**

A factor of nine — and the validation split also selects the checkpoint.

Methodological takeaway

A generalization-gap argument built on the split that performs model selection is not evidence about generalization.

Result 3: H2 supported at $K = 8$, H3 not supported



H2 — capacity dilution. $K = 8$ is the study's only consistent held-out effect: test loss +0.0203, **3/3** seeds, $p = 0.014$. Block-12 entropy also falls in 3/3.

H3 — entropy collapse. The four arms lie on top of one another. Averaged over 12 blocks: +0.020, +0.001, -0.025 bits. None significant.

Patch-norm outlier rate: 1.227%, 1.246%, 1.250%, 1.254%. It rises slightly with K — the opposite of the artifact-removal prediction.

Regime mismatch.

- ▶ Registers were a *pretraining-time* intervention at foundation scale.
- ▶ Ours are randomly initialised into an ImageNet-pretrained backbone and given 50 epochs over 9,000 images.
- ▶ The measured entropy profile of every register arm is the *pretrained* profile — the routing never reorganised.

The artifacts are real but unmoved.

- ▶ Control outlier rate $1.23\% \approx 9\times$ the Gaussian 3σ rate of 0.13% : heavy-tailed structure is present.
- ▶ Registers do not absorb it here.

And $K = 8$ is not free. Eight content-free keys in a 205-token sequence at $d = 192$ measurably cost held-out loss.

Limits we accept

- ▶ $n = 3$, $df = 2$; 15 uncorrected comparisons. Direction count is our primary signal.
- ▶ One architecture, one dataset, one budget.
- ▶ Entropy and outlier rates come from a single 64-image batch — identical across all 12 runs, so pairing is sound, but absolute values are estimates.
- ▶ Entropy averages all query rows, not the [CLS] row alone.

Next experiment

- ▶ Pretrain *with* registers at this scale — the direct test of the mechanism our design cannot reach.
- ▶ [CLS]-restricted entropy as a sharper H3 instrument.
- ▶ Full-split diagnostic pass rather than one batch.

The released sweep runner executes all three unchanged.

1. On a low-data CIFAR-100 ViT-Tiny, **no register configuration improves held-out accuracy** over the register-free control ($75.19 \pm 0.24\%$).
2. The regularization signature appears on validation ($K = 4$: 3/3 seeds, $p = 0.024$) and **does not transfer** to the test split ($p = 0.823$). Training loss is unchanged, so nothing was regularized.
3. $K = 8$ **costs accuracy and held-out loss** (3/3 seeds, $p = 0.014$), consistent with capacity dilution at $d = 192$.
4. No evidence for entropy-collapse prevention in either the entropy or the outlier readout.

Every number here is generated from `outputs/sweep_summary.json` by script. Code, sweep runner and figure generators: github.com/shahin1717/vit-research